BUSINESS REQUIREMENTS DOCUMENT

Byte & Bite

Merchant intelligence for Stellenbosch cafés and quick-service vendors, built on anonymised BoschCard data

Item	Detail
Document	Business Requirements Document (BRD), Draft v2
Developer group	The Final Commit
Product owner	Byte & Bite (Group 4)
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Supersedes	BRD Draft v1 (22 September 2026)
Sources	Client sessions of 22 Sep, 29 Sep and 6 Oct 2026; BRD Draft v1; updated ERD and class diagram (group version, contextual event source)
Status	Draft for group review. Items marked TBC need client confirmation (see Section 15).

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1. Purpose

Byte & Bite converts anonymised BoschCard transaction data into demand, promotion and benchmarking insight for student-oriented cafés and quick-service businesses in Stellenbosch, without exposing individual students or naming competitors.

The platform is a subscription analytics, forecasting and decision-support tool. It shows demand cycles, identifies peak and quiet trading windows, estimates near-term demand, benchmarks performance against similar businesses, and warns vendors when a change in sales coincides with external events, so correlation is never presented as causation. Student privacy and competitor confidentiality are protected throughout.

The guiding idea from the client: **data for better decisions, not intuition.** A successful product is reliable, private, matches the given database constraints, is presented clearly and attractively, and lets vendors make better-informed decisions by comparing themselves with what comparable vendors are doing.

2. Scope

2.1 In scope

Priority	Feature	Notes
MVP (non-negotiable)	Rush Hour Radar	Zone map and weekly heatmap of busy and quiet periods, with automatic discount / no-discount recommendations
MVP (non-negotiable)	Demand Forecast	Weekly horizon, trend / seasonal-average model, one graph with three lines
Principal	Fair Comparison Score	Aggregated peer benchmarking by vendor type and size
Principal	Honest Results Checker	Notification bell plus hover context; flags coinciding events, never causation
Supporting	Dashboard	Quick-glance landing page
Supporting	Weekly report export	Typst-based downloadable weekly report
Supporting	Access management	Invite-only accounts, three roles, location switcher
Supporting	Contextual event administration	Admin maintains exams, sport, holidays, payday and similar events

2.2 Out of scope (v1)

- Manual promotion entry. Managers decide from the data; the system does not record promotions.
- Self-registration. All accounts are created by invitation.
- Machine-learning forecasting. A trend / seasonal-average heuristic is used instead.
- Forecast accuracy targets. Deferred by the client.
- Any new entity that changes the four required ERD entities (Student, Vendor, Vendor Type, Transaction).

3. Users and Roles

There are three roles. Byte & Bite acts as Admin and sees every subscribing vendor; vendor-side users are Managers and Staff.

Role	Description
Admin	Byte & Bite administrator. Full access across all subscribed vendors. Invites users and assigns roles, changes or removes users, edits vendor profile details, and creates, updates and deletes contextual events.
Manager	Vendor decision-maker (up to two per vendor was the client's initial expectation). Views Rush Hour Radar, Demand Forecast, Fair Comparison Score and notifications; exports weekly reports; switches between their own linked vendor locations. Read-only for vendor details: requests changes through the Admin.
Staff	Read-only user for a single vendor. Sees Rush Hour Radar and a reduced Demand Forecast (no revenue) and traffic-related contextual notes only. No Fair Comparison Score and no report export.

Note on Manager vs Staff. Because the platform is otherwise read-only, the Manager / Staff difference is mainly one of **visibility** (which screens and data fields each sees) plus export and location-switching rights, not of editing rights. Draft v1 listed "create, edit, delete" for Managers; this has been removed to match the client's decision that only the Admin edits vendor details.

TBC: whether BoschCard itself receives a separate premium / super-user role for campus-level insight, or uses a standard Admin account.

4. Roles and Permissions Matrix

Feature / screen	Admin	Manager	Staff
Rush Hour Radar (map, heatmap, recommendations)	View	View	View
Demand Forecast	View	View	View, revenue line removed
Fair Comparison Score	View	View	No access (tab hidden, server returns 403)
Honest Results Checker: notification bell	View	View	Contextual notes on traffic data only; no revenue-tied notes
Honest Results Checker: hover context on charts	View	View	Traffic-related context only
Contextual events (exams, sport, paydays, holidays)	Create, update, delete	View only	View only
Export weekly report	Yes	Yes	No
Switch between linked vendor locations	Yes, all vendors	Yes, own linked vendors	Single vendor only
Invite new users	Yes	No	No
Change another user's role / remove a user	Yes	No	No
Reset a locked user's password / revoke access	Yes	No	No
Vendor profile settings (name, address, category)	Edit	View only; sends change requests to Admin	View only

Feature / screen	Admin	Manager	Staff
Change own password	Yes	Yes	Yes

Enforcement. Role checks happen on the server, not only in the interface. A Staff user cannot retrieve restricted data by inspecting network requests, because the server withholds it: the Fair Comparison endpoint returns 403 for Staff, and the expected-revenue field is stripped from the forecast response (and any note that would reveal revenue is never sent) before it leaves the server.

5. Key Features

5.1 Rush Hour Radar (MVP)

The main landing analytics view. It identifies both busy and quiet periods.

- **Zone map.** A geographical map of Stellenbosch divided into zones. Each zone is classified as quiet, moderate, busy or very busy based on the vendors inside it. Quartile cut-offs (25 / 50 / 75) drive the classes, but only the labels are shown, never the percentages.
- **Filters.** By vendor type and by operational size (small, medium, large), where size is derived from transaction volume. Colour gradient to show intensity.
- **Weekly heatmap.** Day of week by hour, with visual charts and a vendor-type filter. Daily-level history is available.
- **Baseline and thresholds.** Compared with the same day and hour over the previous four weeks. A period is quiet when it is about 30% below baseline and busy when about 20% above (**TBC**, see Section 15).
- **Recommendations.** When a recurring quiet period crosses the threshold, the system generates a suggested discount, informed by how past discounts performed (Transaction.discount against sales). During busy periods it advises no discount. A recommendation is a suggestion only; the Manager chooses whether to act.
- **Next-step guidance.** Extension requested in Session 3: show the Manager how last week went, where they did well, and the factors to apply this week to reach a high point.
- **Location privacy.** Individual vendor locations are generalised (zone level, coordinates reduced) so competitors cannot be pinpointed.

5.2 Demand Forecast (MVP)

- Weekly horizon (next 7 days), with daily historical data available for context.
- Trend / seasonal-average model; no machine learning.
- One line graph with three lines: **Expected Transactions, Expected Customers, Expected Revenue.**
- A fourth measure (average order value, or average spend per student) was proposed. It is **undecided**; the client will come back to us. If added, spend-per-student values based on too few students are masked.
- Values are shown as estimates, not guaranteed figures.
- Hovering a point shows the values for that day and any coinciding contextual events.
- Staff see the graph without the revenue line.

5.3 Fair Comparison Score

- Peers are grouped by **vendor type + size** (small cafés with small cafés). Size is computed from transaction volume.
- A comparison is only shown when at least **3** peer businesses participate. This prevents reverse-engineering of an individual competitor.
- Result is a label (above average / average / below average), not a percentile. First view shown to a Manager: their own figure, the aggregated peer average, the result, and the peer group description (for example, "Small cafés, 8 participating businesses").
- Only aggregated statistics are shown; never individual competitor names or volumes.
- Helps a Manager tell an isolated problem from a campus-wide pattern (if everyone has a bad week, no action is needed).
- Filterable by day. Not available to Staff.

5.4 Honest Results Checker

- **Trigger.** A notification is created when sales shift by about 30% up or down against the normal baseline.
- **Notification bell.** Opens a list of notifications; selecting one shows a detailed breakdown. The message states only that unusual volume **coincided** with events (for example, a rugby fixture). It never claims causation. Notifications are persistent and can be read, pinned and cleared.
- **Hover context.** Chart points show context directly (for example, "Sales up 15%; a Maties rugby event took place the same day").
- **Contextual events** include exams, recess, holidays, graduation, sport (Maties and SA rugby), orientation and engineering weeks, and payday. Events are general student-wide events, not niche groups.
- The client found flashing flags and pop-ups annoying, so the bell approach is preferred.
- If event data is incomplete, the system shows an "analysis limited to baseline trends" notice rather than guessing.

5.5 Contextual Event Administration

- Admin can create, update and delete events using an **event type dropdown + event name + date (range)**.
- Event types include Exam, Holiday, Sport, Payday, Graduation and similar (final list **TBC**).
- The calendar is seeded from a verified list of 191 Stellenbosch dates (exam weeks, recess, public holidays, graduation, orientation week, major events), with Admin additions on top.
- Events are stored in the **ContextualEvent** table (see Section 10). This replaces the earlier plan to hold the calendar in a flat file, now that the supporting tables sit outside the four original entities.

5.6 Dashboard and Navigation

- A quick-glance landing page after login (client left detailed content to our discretion). Proposed from the wireframe: revenue, transactions, average basket, peak hour, weekly trend, quick analytics and the top recommendation.
- Main navigation: Dashboard, Rush Hour Radar, Demand Forecast, Fair Comparison, Context Checker (notifications), Reports, Locations, Settings.

- Location switcher dropdown; log-out control in the top-right.

5.7 Reporting

- Weekly report exported with Typst, downloadable by Manager and Admin only.
- Export button available on each of the three main feature pages. Report contents are at our discretion; proposed sections are weekly peak and quiet summary, forecast, fair comparison standing, and context notes.
- Each export is logged (ReportExport table).

5.8 Authentication

- **Invite-only.** Admin invites by email and assigns the role. The invitation link confirms the email address and lets the user set their own password; it expires after **24 hours**. An email service (Gmail was suggested) is to be chosen.
- **Failed login.** A simple error message ("incorrect email or password"). After repeated failures the account is flagged and the Admin sees a pop-up with two choices: revoke access or send a password reset.
- **Forgot password.** The user is told an email was sent and taken to a page that returns them to login. Mechanism (email reset link or Admin reset) is **TBC**.
- **Landing page.** The user lands on a quick-glance dashboard.

6. Business Rules and Thresholds

Rule	Value / decision	Source
Minimum students before time-slot data is shown	5 (reduced from 10 in Session 3; client was shown Options 1 to 3 and notes record "drop it to 5"). Below this, data is hidden. Option 3 (keep 10, use 10 weeks of history) was our recommendation. Confirm final choice.	Session 1, Session 3
Minimum businesses in a comparison group	3	Session 1, NFR-02
Small groups	A result with only one student/business is merged into another group	Session 1
Vendor size	Small / medium / large, derived from transaction volume (and venue capacity where available); size matters more than type	Sessions 1, 2
Zone classes	Quiet, moderate, busy, very busy at 25 / 50 / 75 cut-offs; labels only	Session 2
Baseline window	4 weeks, same day and hour	Session 2
Quiet / busy flag	About 30% below / 20% above baseline	Session 2
Notification trigger	About 30% increase or decrease	Session 3
Invite validity	24 hours	Session 3
Time slots for privacy checks	3-hour slots	Session 3
Forecast horizon	Weekly (7 days)	Session 1

7. Functional Requirements

Priorities follow MoSCoW. Requirement numbers are new in this version.

ID	Requirement	Priority	Users
Access and accounts			
FR-01	Account creation is invite-only; no public self-registration.	Must	Admin
FR-02	Admin invites a user by email and assigns a role (Manager or Staff). The link lets the user set their own password and expires after 24 hours.	Must	Admin
FR-03	Users log in with email and password. A failed login shows a simple error; repeated failures flag the account and prompt the Admin to revoke access or reset the password.	Must	All
FR-04	Forgot-password flow confirms an email was sent and returns the user to login (mechanism TBC).	Should	All
FR-05	Users can change their own password and log out from the top-right.	Should	All
FR-06	Admin can change a user's role or remove a user.	Should	Admin
FR-07	One account can be linked to several vendor locations; the user switches via a dropdown and sees data for the selected location only.	Should	Admin, Manager
FR-08	Role checks are enforced server-side for every request.	Must	System
Rush Hour Radar			
FR-09	Show a zoned map of Stellenbosch with each zone labelled quiet, moderate, busy or very busy.	Must	All
FR-10	Filter by vendor type and operational size; colour gradient; pan and zoom including two-finger gestures; enlarged dots / areas.	Must	All
FR-11	Show a day-of-week by hour heatmap with daily historical data, filterable by vendor type.	Must	All
FR-12	Classify periods against a four-week same-day, same-hour baseline.	Must	System
FR-13	When a quiet period recurs, auto-generate a suggested discount based on past discount performance; during busy periods advise no discount. Recommendations are suggestions only.	Must	Manager
FR-14	Provide last-week review and suggested actions for the coming week.	Could	Manager
FR-15	Suppress any slot or group below the privacy threshold and show an "insufficient data" message.	Must	System
Demand Forecast			
FR-16	Display a 7-day forecast using a trend / seasonal-average method, as one graph with three lines: expected transactions, customers and revenue.	Must	All
FR-17	Label forecast values as estimates; hover shows values and coinciding contextual events.	Must	All
FR-18	For Staff, strip expected revenue from the response server-side.	Must	Staff
Fair Comparison Score			
FR-19	Compare a vendor with an aggregated peer group (same type and size); require at least 3 peers; show above / average / below, peer average and group description.	Should	Admin, Manager

ID	Requirement	Priority	Users
FR-20	Allow the comparison to be filtered by day. Never reveal competitor identities or individual values.	Should	Admin, Manager
Honest Results Checker and events			
FR-21	Generate a notification when sales shift by about 30% against baseline and an event overlaps.	Should	System
FR-22	Notification bell lists notifications; selecting one shows a detailed breakdown. Notifications persist and can be marked read, pinned and cleared.	Should	Admin, Manager
FR-23	Wording flags coincidence only, never causation.	Must	System
FR-24	Hover on chart points shows contextual details.	Should	All
FR-25	Staff receive contextual notes on traffic data only, never revenue-related notes.	Should	Staff
FR-26	Admin creates, updates and deletes contextual events (type dropdown, name, dates); the calendar is seeded from the verified 191-date list.	Should	Admin
FR-27	If event data is unavailable, fall back to historical day-of-week averages and signal baseline mode.	Could	System
Reporting, dashboard and data			
FR-28	Export a weekly report (Typst) from each main feature page; Manager and Admin only; log each export.	Should	Admin, Manager
FR-29	Landing dashboard with quick-glance indicators and the top recommendation.	Should	All
FR-30	Ingest BoschCard transactions (timestamp, value, discount, vendor) and randomise student identifiers; no student name, ID number or address reaches the pipeline output or any client payload.	Must	System
FR-31	Vendor locations are generalised to zone level in any client-visible output.	Must	System
FR-32	Admin edits vendor details; Managers send change requests to the Admin.	Should	Admin, Manager

8. Non-Functional Requirements

ID	Requirement
NFR-01 Privacy and POPIA	No individual student spending profile or identifying card sequence may be exposed or queryable through any dashboard or client payload. Student identifiers are randomised and never shown downstream. Slots below the privacy threshold are hidden. No student ID, name or address leaves the data pipeline.
NFR-02 Competitor confidentiality	Comparison cohorts need at least 3 businesses. Competitor names, specific volumes and turnover are never shown.
NFR-03 Performance	30-day heatmaps and forecast charts render in 2.0 seconds or less. Daily batch ingestion must not degrade dashboard read latency.
NFR-04 Display and interaction	Web application with hover-interactive charts and filters; readable fonts (larger font on the forecast view) and good logo and sidebar contrast. Responsive for laptops, tablets and mobile browsers.

ID	Requirement
NFR-05 Architecture	Built on the provided ERD (Student, Vendor, Vendor Type, Transaction), kept intact. Supporting tables are added alongside, with no links back into the original four (Section 10).
NFR-06 Security	Server-side role enforcement; hashed passwords and invite tokens; account lockout flag; 24-hour invite expiry.
NFR-07 Reliability	Graceful degradation to day-of-week averages if event data is unavailable, with the baseline mode clearly signalled.
NFR-08 Supportability	Separate the aggregation engine, context scoring module and reporting UI so machine learning can be added later without breaking contracts.
NFR-09 Clarity	Visually distinguish facts (historical transactions), estimates (forecasts) and warnings (context notes).

9. FURPS+ Summary

Category	Summary
Functionality	Anonymised ingestion; Rush Hour Radar with quiet-period triggers; Demand Forecast; Fair Comparison Score; Honest Results Checker; multi-location management with role-based access.
Usability	Quick-glance dashboard, clear hierarchy of facts, estimates and warnings, hover details, filters, responsive layout.
Reliability	Fallback to historical averages; suppression of low-volume data rather than unreliable output.
Performance	Heatmaps and forecasts in 2 seconds or less; batch ingestion without read slowdown.
Supportability	Modular engines; ML-ready contracts.
+ Constraints	POPIA alignment; commercial confidentiality; no self-registration, manual promotions or ML in v1; four original ERD entities unchanged; real relational database (not a spreadsheet).

10. Data Model

The data model keeps the four original entities exactly as supplied and adds five supporting tables for access, notifications, exports and contextual events. In both diagrams the **green** group is the original model and the **maroon** group is new. There are deliberately no foreign-key links between the two groups.

10.1 Entity relationship diagram (SEE NEW ERD AT BOTTOM OF DOCUMENT)

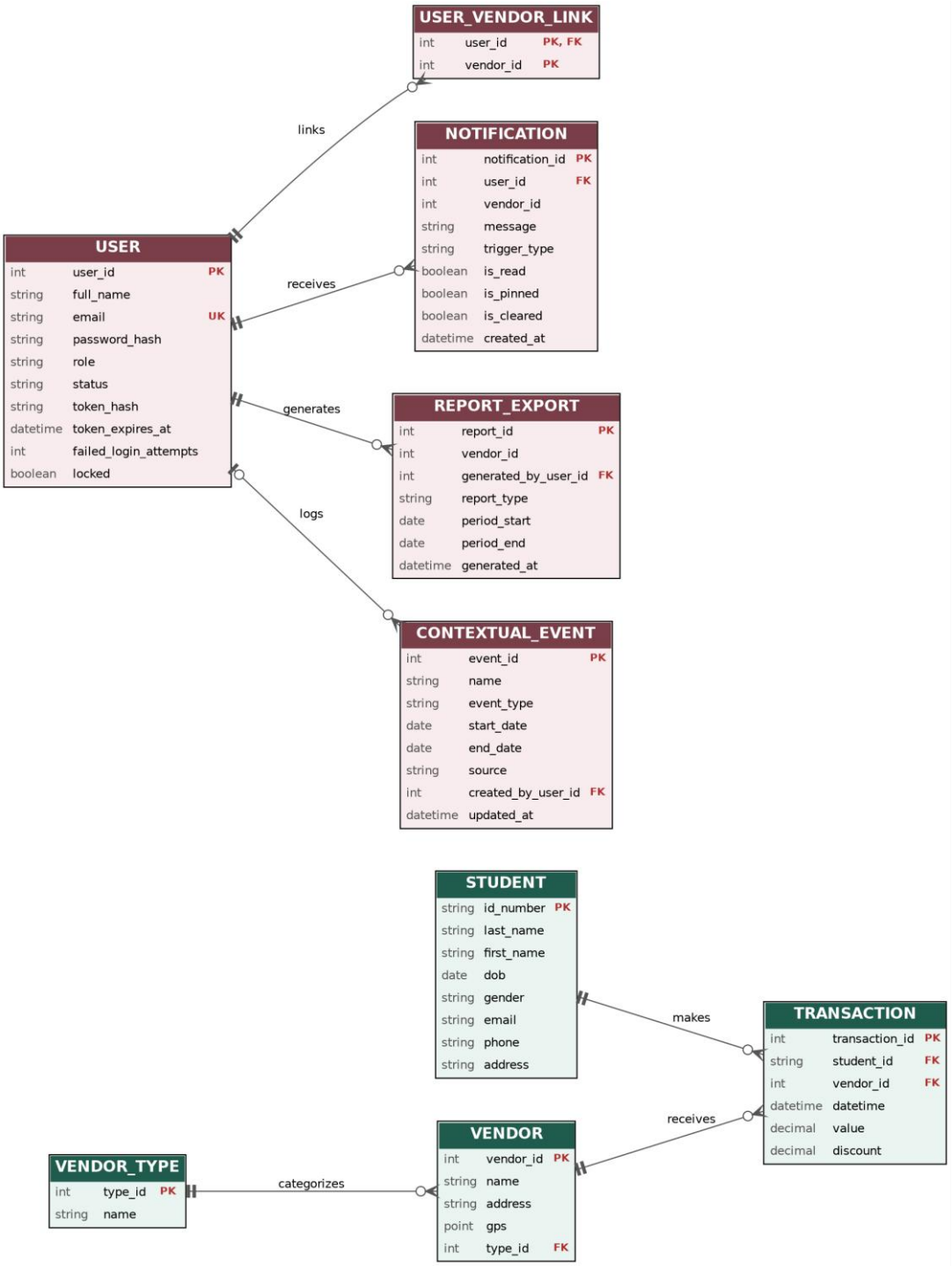


Figure 1: ERD, original four tables plus supporting tables

10.2 Class diagram (SEE NEW CLASS DIAGRAM AT BOTTOM OF DOCUMENT)

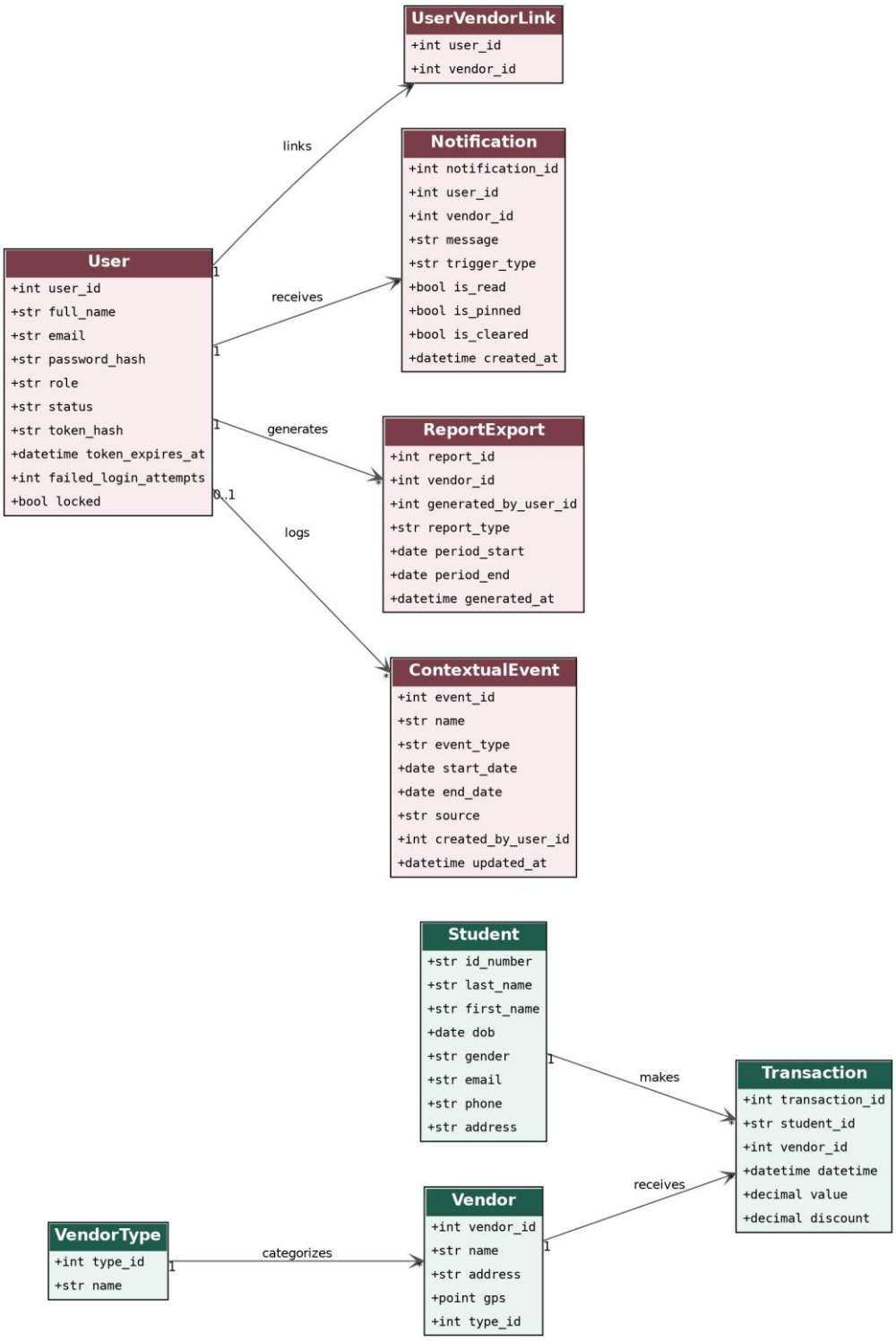


Figure 2: Class diagram matching the ERD

10.3 Relationships

From	To	Cardinality	Meaning
Student	Transaction	1 to many	A student makes many transactions
Vendor	Transaction	1 to many	A vendor receives many transactions
Vendor Type	Vendor	1 to many	A type categorises many vendors

From	To	Cardinality	Meaning
User	User Vendor Link	1 to many	A user is linked to one or more vendors
User	Notification	1 to many	A user receives many notifications
User	Report Export	1 to many	A user generates many report exports
User	Contextual Event	0..1 to many	An event may be logged by a user (blank for seeded events)

10.4 Data dictionary

Original entities (unchanged)

Student

Field	Type	Key	Notes
id_number	string	PK	Contains personal data; must be randomised in the pipeline and never exposed
last_name, first_name	string		Personal data; never leaves the pipeline
dob, gender	date, string		
email, phone, address	string		Personal data; never leaves the pipeline

Vendor Type

Field	Type	Key	Notes
type_id	int	PK	
name	string		For example café or restaurant; grouping depends more on size than type

Vendor

Field	Type	Key	Notes
vendor_id	int	PK	
name	string		
address	string		Not shown to other vendors
gps	point		Generalised to zone level for display
type_id	int	FK	References Vendor Type

Transaction

Field	Type	Key	Notes
transaction_id	int	PK	
student_id	string	FK	References Student; randomised downstream
vendor_id	int	FK	References Vendor

Field	Type	Key	Notes
datetime	datetime		Basis of hourly and day-of-week analysis
value	decimal		Spend; basis of revenue
discount	decimal		Basis of discount performance analysis

New supporting entities

User

Field	Type	Key	Notes
user_id	int	PK	
full_name	string		
email	string	UK	Login identifier; unique
password_hash	string		Hashed; never plain text
role	string		Admin, Manager or Staff
status	string		Account state, for example invited, active or revoked (values to be fixed in design)
token_hash	string		Hashed invite / reset token
token_expires_at	datetime		Invite link expiry (24 hours)
failed_login_attempts	int		Counts failed logins
locked	boolean		Set after repeated failures; Admin revokes or resets

User Vendor Link

Field	Type	Key	Notes
user_id	int	PK, FK	References User
vendor_id	int	PK	Vendor the user may access (see note below)

Notification

Field	Type	Key	Notes
notification_id	int	PK	
user_id	int	FK	Recipient
vendor_id	int		Vendor the notice relates to
message	string		Coincidence wording only
trigger_type	string		What caused it, for example a 30% shift
is_read, is_pinned, is_cleared	boolean		Support read, pin and clear
created_at	datetime		

Report Export

Field	Type	Key	Notes
report_id	int	PK	
vendor_id	int		Vendor the report covers
generated_by_user_id	int	FK	References User
report_type	string		For example weekly
period_start, period_end	date		Reporting period
generated_at	datetime		

Contextual Event

Field	Type	Key	Notes
event_id	int	PK	
name	string		For example Maties rugby, graduation
event_type	string		Exam, Holiday, Sport, Payday, Graduation and so on
start_date, end_date	date		Single-day events use the same date
source	string		Records where the event came from (seeded calendar or Admin entry)
created_by_user_id	int	FK	Admin who logged it; empty for seeded events
updated_at	datetime		

10.5 Design notes

- **Original model untouched.** The lecturer controls the original ERD, so the new tables stand alone. Vendor references in User Vendor Link, Notification and Report Export are plain integers with no database foreign key to Vendor. Their integrity must therefore be enforced in the application. This is an accepted trade-off and should be stated in the design justification.
- **Contextual events.** The calendar now lives in a small supporting table instead of a flat file. The lecturer's feedback allows extra tables as long as the original four are unchanged. Admin upload of a spreadsheet remains an optional way to load many dates at once (**TBC**).
- **Derived values, not stored.** Operational size, zone, baseline, busy / quiet class, forecast and peer averages are calculated from Transaction and Vendor data rather than stored.
- **Promotions.** Promotions are not recorded, per the client. Discount performance comes from Transaction.discount.
- **Privacy.** Student personal fields exist in the supplied model but must never reach any output or client payload.

11. Use Cases

The use case diagram from Draft v1 is retained and should be updated to match the list below. Applying a recommendation extends viewing the Rush Hour Radar; viewing any analytics screen includes logging in.

ID	Use case	Actors	Notes
UC-01	Log in	Admin, Manager, Staff	Included by all viewing use cases
UC-02	Set password from invite link	Invited user	24-hour link
UC-03	Request forgot-password	All	Mechanism TBC
UC-04	Invite new user	Admin	
UC-05	Change user role / remove user	Admin	
UC-06	Review failed-login alert; revoke or reset	Admin	
UC-07	Manage vendor profile settings	Admin	Managers send requests
UC-08	Enter / update / delete contextual event	Admin	
UC-09	View Rush Hour Radar	Admin, Manager, Staff	
UC-10	View recommendation and acknowledge or dismiss it	Manager	Extends UC-09; not a promotion record
UC-11	View Demand Forecast	Admin, Manager, Staff	Staff without revenue
UC-12	View Fair Comparison Score	Admin, Manager	
UC-13	View notifications; pin or clear	Admin, Manager	Staff see contextual notes only
UC-14	Hover for Honest Results context	All	
UC-15	Export weekly report	Admin, Manager	
UC-16	Switch vendor location	Admin, Manager	
UC-17	Change own password	All	

Activity flows from v1 retained: (1) Evaluate Promotional Outcome via the Honest Results Checker (query transactions and context registry, flag confounding events, show warning with cohort movement); (2) Authenticate, Resolve Access and Enable Analytics (credential check, role resolution, location selection, role-based feature set).

12. User Stories

Duplicate identifiers in v1 (two US03 and two US05) have been corrected.

ID	User story	Key acceptance criteria	Priority
US01	As a Manager, I want to view the Rush Hour Radar so I can identify recurring busy and quiet trading periods.	Zone map with four classes; day-by-hour heatmap; filters; history; suggestion generated when threshold met.	Must

ID	User story	Key acceptance criteria	Priority
US02	As a Manager, I want the Demand Forecast so I can anticipate demand and plan operations.	7-day horizon; no ML; three lines on one graph; shown as estimates.	Must
US03	As a Manager, I want recommendations for discounts or no discounts so I can make informed promotional decisions.	Based on recurring patterns and past discount performance; reasoning shown; suggestion only.	Must
US04	As a Manager, I want to compare my business with similar businesses so I know whether I am above or below my peers.	Group by type and size; at least 3 peers; aggregated only; label result.	Should
US05	As a Manager, I want contextual notifications when events overlap unusual activity so I do not wrongly credit a promotion.	About 30% shift trigger; coincidence wording only; bell list; pin and clear.	Should
US06	As a Manager, I want to switch between linked locations so I can see each one's analytics from one account.	Dropdown; data for selected location only.	Should
US07	As a Manager, I want to export a weekly report so I can use insights outside the platform.	Typst weekly report; key analytics; downloadable.	Should
US08	As a Staff member, I want to view operational analytics so I understand current and historical activity.	Radar and traffic forecast only; no revenue; no Fair Comparison; no export.	Should
US09	As an Admin, I want to invite users and assign roles so access is controlled without self-registration.	Email invite; 24-hour link; role assigned; password set by user.	Must
US10	As an Admin, I want to manage contextual events so relevant events are considered in analysis.	Create, update, delete; type dropdown; used by analysis and notifications.	Should
US11	As an Admin, I want to handle failed-login alerts so locked users can be revoked or reset.	Pop-up with revoke and reset options.	Should
US12	As an Admin, I want to edit vendor details so records stay correct when Managers request changes.	Only Admin can edit; Manager view-only.	Should

13. Technology Direction

No stack is mandated, but the choice must be justified in writing and run against the given ERD in a real database.

Layer	Proposed choice	Reason
Database	PostgreSQL	Fits the relational ERD, proper foreign keys and joins for aggregation
Backend	Python with FastAPI	Low setup overhead, well documented, gentle learning curve for a mixed-experience group

Layer	Proposed choice	Reason
Frontend	React / JavaScript	Accepted by the client; supports interactive charts, maps and filters
Reporting	Typst	Client-requested weekly report export
Email	To be chosen (Gmail suggested)	Invitations and password resets

The final choice and justification (team skills, ERD fit, setup overhead, six-week timeline) is documented separately.

14. Client Feedback on MVP Version 1

- Map: support scrolling and two-finger gestures; make dots / areas bigger.
- Demand forecast: increase font size.
- Branding: lighter logo or better contrast for side tab and logo.
- Top-right control should log out.
- Landing page should be a quick glance.
- Notifications: trigger at a 30% shift; pin and clear options; recurring occurrences may live on a separate page.
- Fair Comparison: add a day filter.
- Show how the demand forecast looks, then compare against wireframes.
- Technical: create database tables; they need not be BoschCard data.

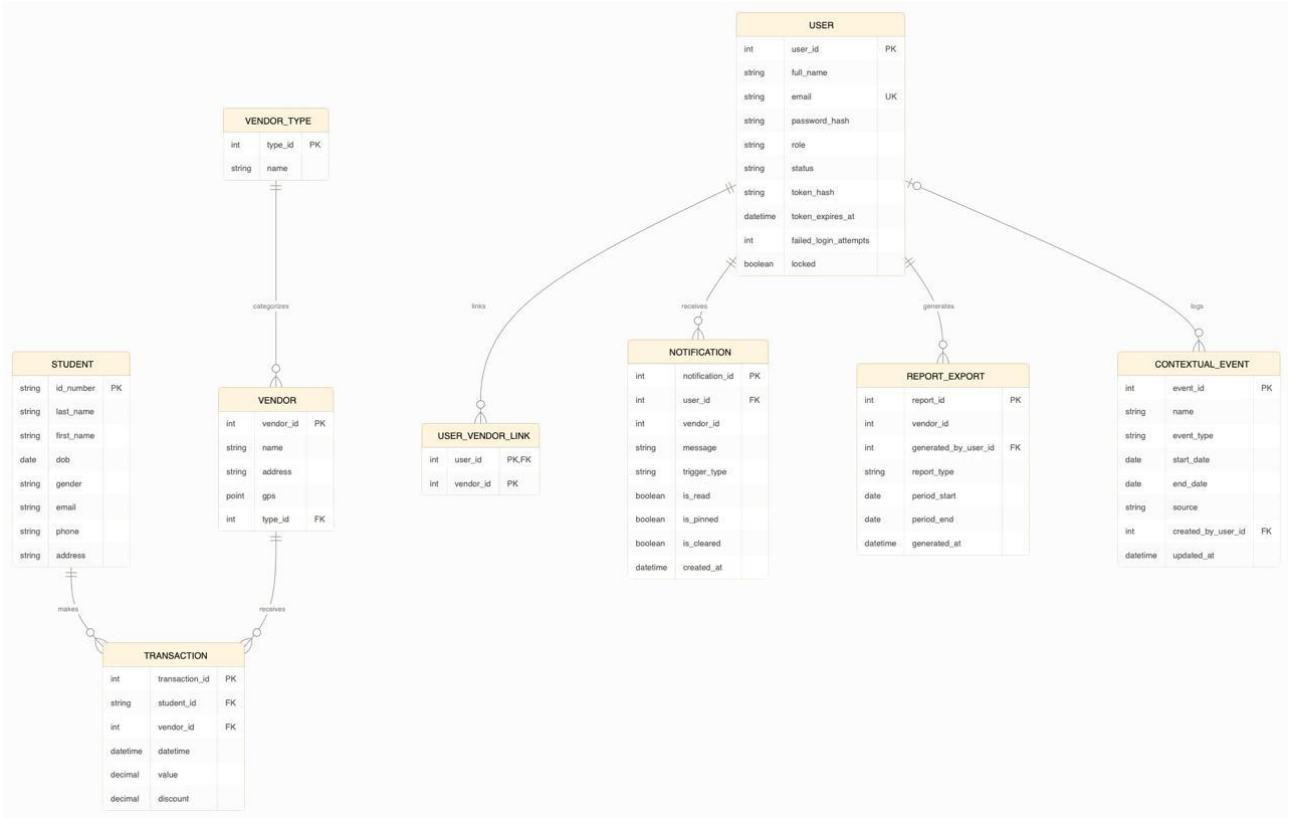
15. Open Items and Decisions to Confirm

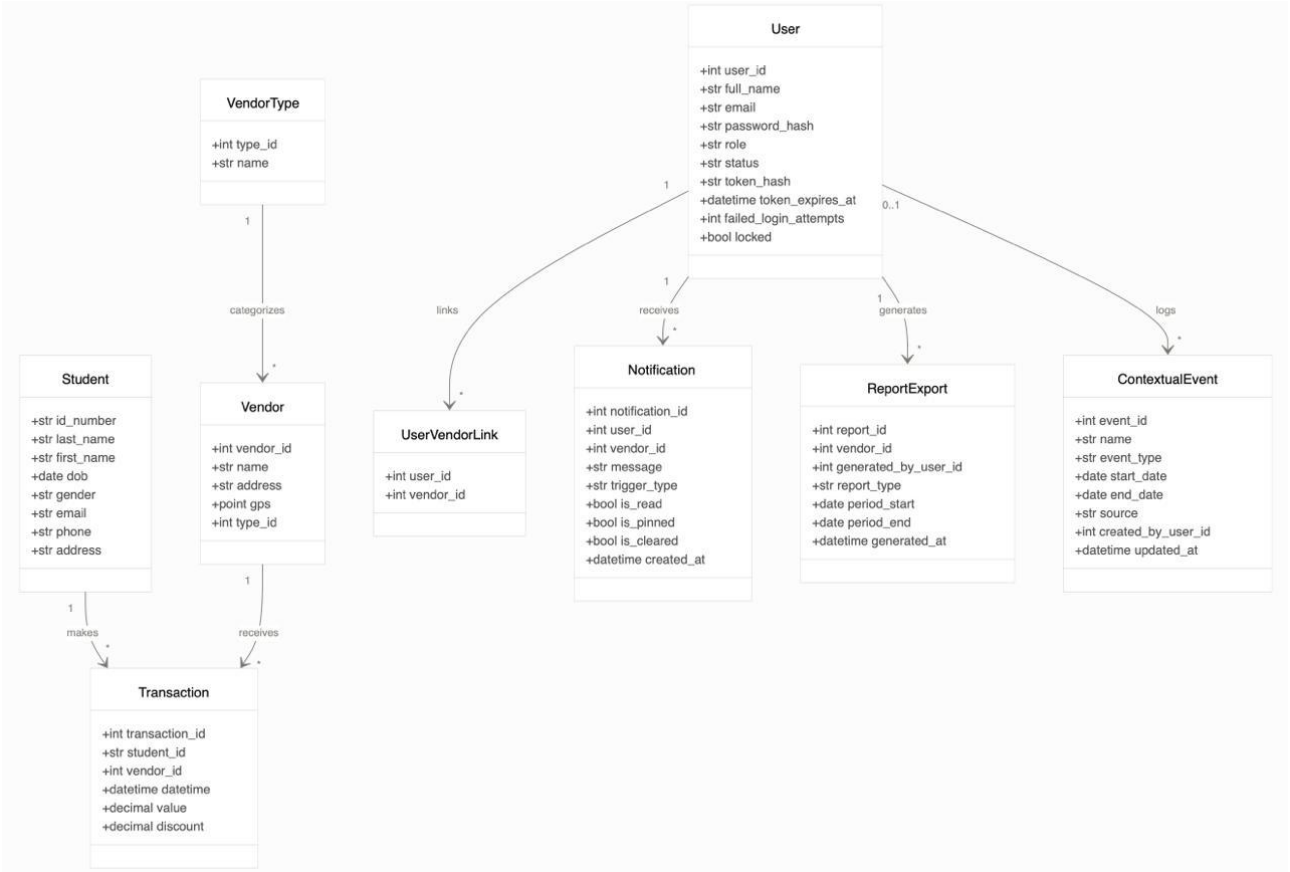
#	Item	Detail
1	Privacy threshold	Confirm 5 students (Option 1) versus keeping 10 with ten weeks of history (our recommended Option 3).
2	Recurring definition	Session 2: three of the last four same-weekday occurrences. The wireframe shows eight of twelve weeks. Choose one.
3	Quiet and busy percentages	Minutes read "30% below + 20% above baseline". Confirm busy is 20% above.
4	Time granularity	Heatmap is hourly by day; privacy checks use 3-hour slots. Confirm how both are shown.
5	Fourth forecast measure	Average order value, average spend per student, or none. Client to confirm.
6	Forecast period	Next 7 days from latest data or from the current date.
7	Forgot password	Email reset link or Admin reset.
8	Notification display	Client dislikes pop-ups; Session 3 notes mention a pop-up for regular occurrences. Confirm bell only, or an in-page banner on a separate page.
9	Notification retention	Remain after being read, or disappear. The data model supports retention.
10	BoschCard role	Separate premium / super-user role, or standard Admin.
11	Event types and upload	Final dropdown list, and whether Admin bulk-upload is needed alongside the form.
12	Weekly report contents and export scope	Left to our discretion; confirm sections and per-page export.

#	Item	Detail
13	Dashboard contents	Left to our discretion; confirm the quick-glance items.
14	Number of Managers per vendor	Two Managers and one read-only user were mentioned in Session 1.
15	Email service	Choose and set up for invites and resets.
16	Synthetic dataset	Delivery date. This blocks development.

16. Changes from BRD Draft v1

Area	Change
Roles	Manager "create, edit, delete" removed; Admin is the only editor of vendor details and events.
Staff access	Fair Comparison hidden (403); revenue removed from forecast server-side; export denied; contextual notes limited to traffic.
Privacy threshold	Reduced from 10 students to 5 after Session 3 discussion; final choice flagged.
Contextual events	Moved from a flat file (Session 3 prep) to the new ContextualEvent table, seeded with 191 verified dates.
Rush Hour Radar	Now a zoned geographical map with four classes, plus the heatmap, four-week baseline and dynamic discount recommendations.
Honest Results Checker	Bell and hover together; 30% trigger; pin and clear; coincidence wording only.
Authentication	Invite link valid 24 hours; failed-login handling; forgot-password flow added.
Data model	Five supporting tables added (User, User Vendor Link, Notification, Report Export, Contextual Event); original four unchanged.
User stories	Duplicate IDs fixed; stories added for failed-login handling and vendor edits.


USER
int user_id PK
string full_name
string email UK
string password_hash
string role
string status
string token_hash
datetime token_expires_at
int failed_login_attempts
boolean locked
VENDOR_TYPE
int type_id PK
string name
categorizes
links
receives
generates
logs
STUDENT
string id_number PK
string last_name
string first_name
date dob
string gender
string email
string phone
string address
VENDOR
int vendor_id PK
string name
string address
point gps
int type_id FK
USER_VENDOR_LINK
int user_id PK,FK
int vendor_id PK
NOTIFICATION
int notification_id PK
int user_id FK
int vendor_id
string message
string trigger_type
boolean is_read
boolean is_pinned
boolean is_cleared
datetime created_at
REPORT_EXPORT
int report_id PK
int vendor_id
int generated_by_user_id FK
string report_type
date period_start
date period_end
datetime generated_at
CONTEXTUAL_EVENT
int event_id PK
string name
string event_type
date start_date
date end_date
string source
int created_by_user_id FK
datetime updated_at
makes
receives
TRANSACTION
int transaction_id PK
string student_id FK
int vendor_id FK
datetime datetime
decimal value
decimal discount


User
+int user_id
+str full_name
+str email
+str password_hash
+str role
+str status
+str token_hash
+datetime token_expires_at
+int failed_login_attempts
+bool locked
VendorType
+int type_id
+str name
1
categorizes
*
links
receives
generates
logs
0..1
Student
+str id_number
+str last_name
+str first_name
+date dob
+str gender
+str email
+str phone
+str address
Vendor
+int vendor_id
+str name
+str address
+point gps
+int type_id
UserVendorLink
+int user_id
+int vendor_id
Notification
+int notification_id
+int user_id
+int vendor_id
+str message
+str trigger_type
+bool is_read
+bool is_pinned
+bool is_cleared
+datetime created_at
ReportExport
+int report_id
+int vendor_id
+int generated_by_user_id
+str report_type
+date period_start
+date period_end
+datetime generated_at
ContextualEvent
+int event_id
+str name
+str event_type
+date start_date
+date end_date
+str source
+int created_by_user_id
+datetime updated_at
makes
receives
Transaction
+int transaction_id
+str student_id
+int vendor_id
+datetime datetime
+decimal value
+decimal discount